

Huawei ICT Competition 2025 – 2026

Innovation Competition

Beirut Arab University + SYNShield + Early SYN-Flood DDOS Attack Detection



HUAWEI ICT COMPETITION — INNOVATION (REGIONAL) · APRIL 11, 2026

SYNSheild

Huawei MindSpore + ResNeXt-50 for Early SYN-Flood DDoS Detection

Team: Wafik Ibrahim · Mohammed Farhat · Abed Al Rida Nehme **Instructor:**
Ali Nassar · **Beirut Arab University (BAU), Lebanon**





What We'll Cover

We start practical, then move to formal research proof.

Part A

Problem-Value Framing

1. Realistic scenario
2. Why existing solutions fall short
3. SYNShield's core advantage
4. Value for SOC teams

Part B

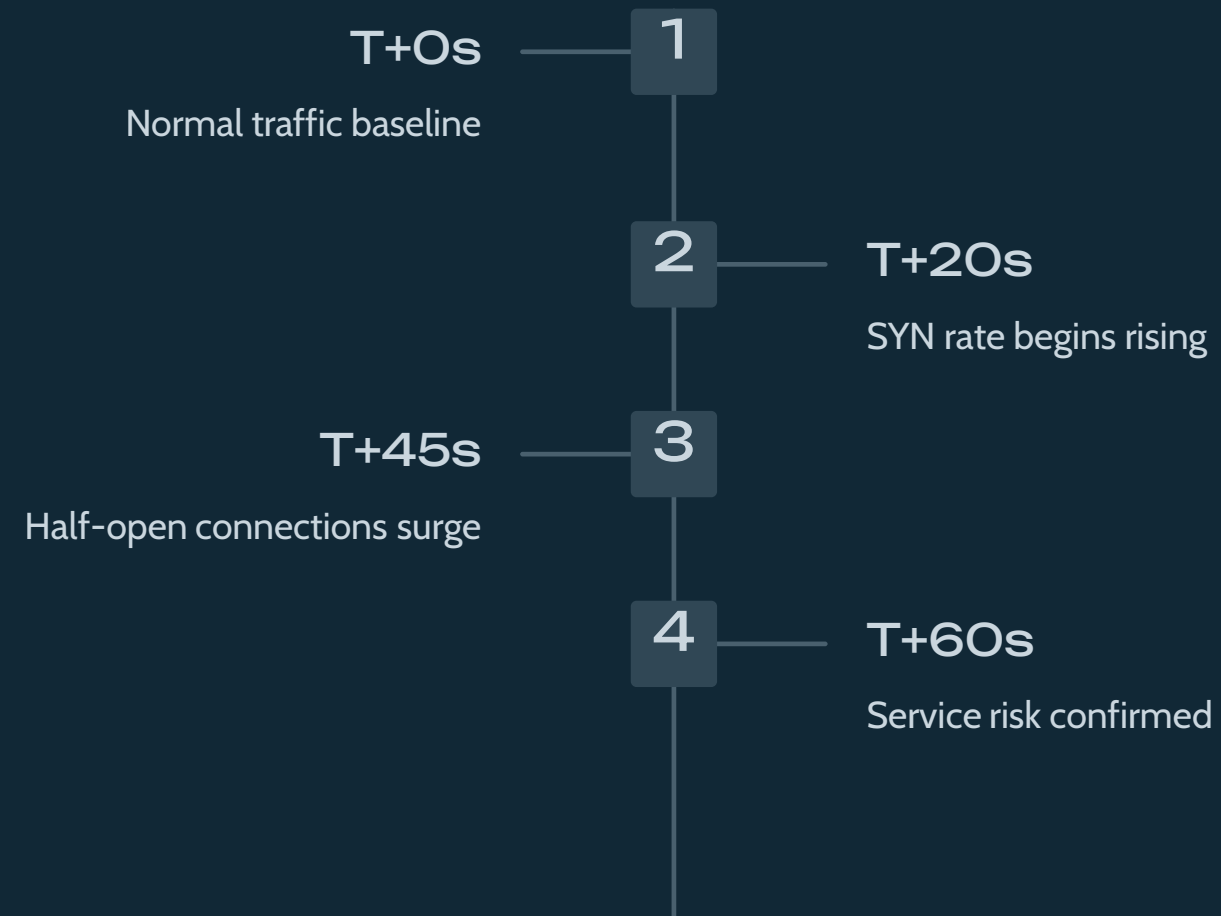
Research Evidence & Deployment Realism

1. Architecture and technical approach
2. Methodology and evaluation framework
3. Dataset design and experimental validation
4. Reproducibility and deployment readiness



A SYN Flood in the First Minute

How fast does an attack become a crisis? Here's the window we're closing.



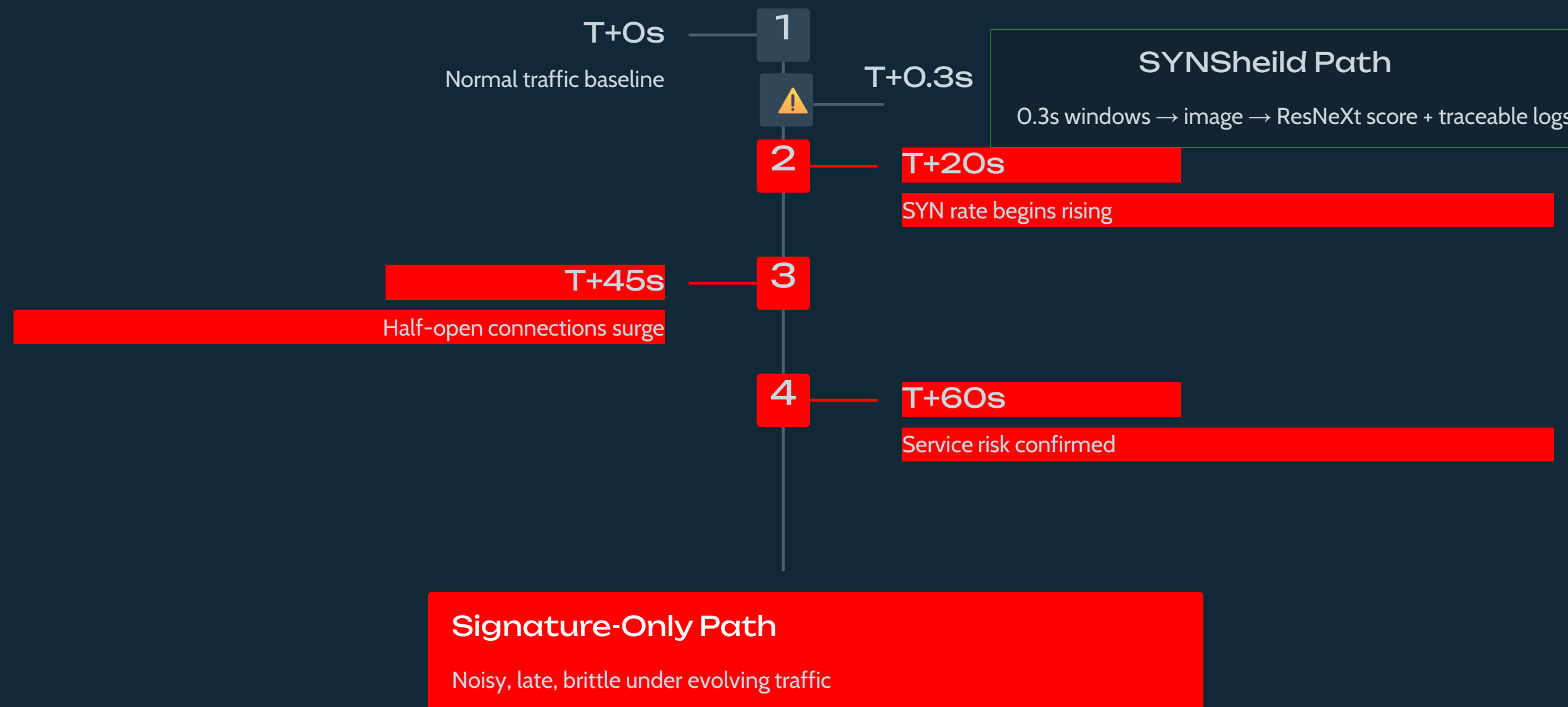
Signature-Only Path

Noisy, late, brittle under evolving traffic



A SYN Flood in the First Minute

How fast does an attack become a crisis? Here's the window we're closing.





Without vs. With SYNSheild

Without SYNSheild

- Brittle signature-only behavior under evolving traffic patterns
- Many prototypes lack blind testing and reproducibility
- Accuracy-only reporting hides real-world failure modes

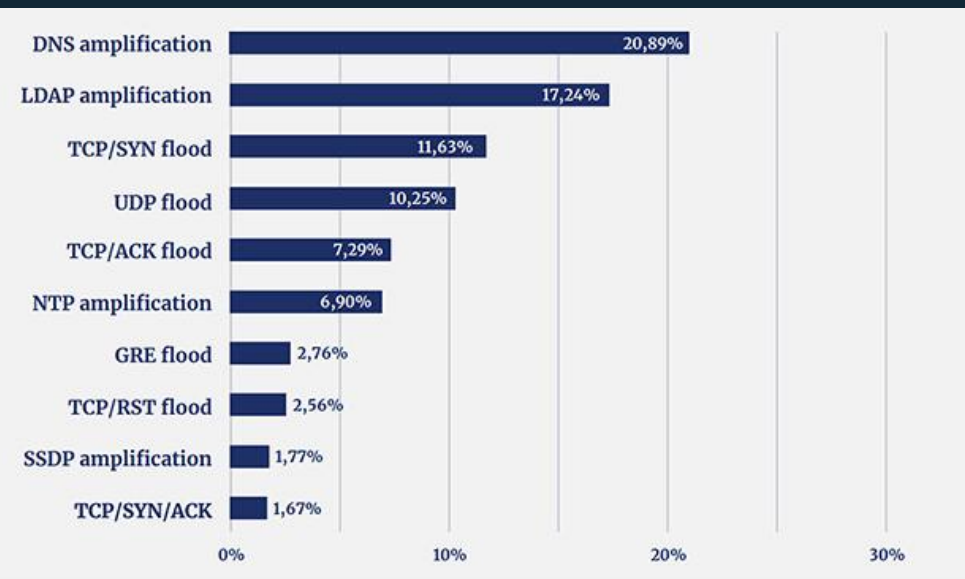
With SYNSheild

- End-to-end pipeline: PCAP/ZIP → preprocessing → evaluation
- Fair metrics (macro-F1), blind tests, reproducible logs
- Huawei-aligned framework path: MindSpore + mindcv

HUAWEI



Background, Motivation & Scope



Distribution of DDoS Attack Types by volume in the year 2024

Background

Protocol-level abuse — especially SYN-flood — continues to rise in operational networks, outpacing signature-based defenses.

Motivation

Enable **earlier, more reliable** SYN-flood detection to support real-time SOC decisions and availability protection.

Scope

In: Binary SYN-flood detection from traffic-window images. **Out:** Multi-attack taxonomy & automated mitigation loop (future phase).



The Problem We're Solving

Detect SYN-flood behavior **early and reliably** — before service availability is compromised.



Class Imbalance

Attack samples are rare; models must learn without bias toward the majority class.



Traffic Domain Shift

Models trained on one dataset often fail when deployed on real, evolving traffic.



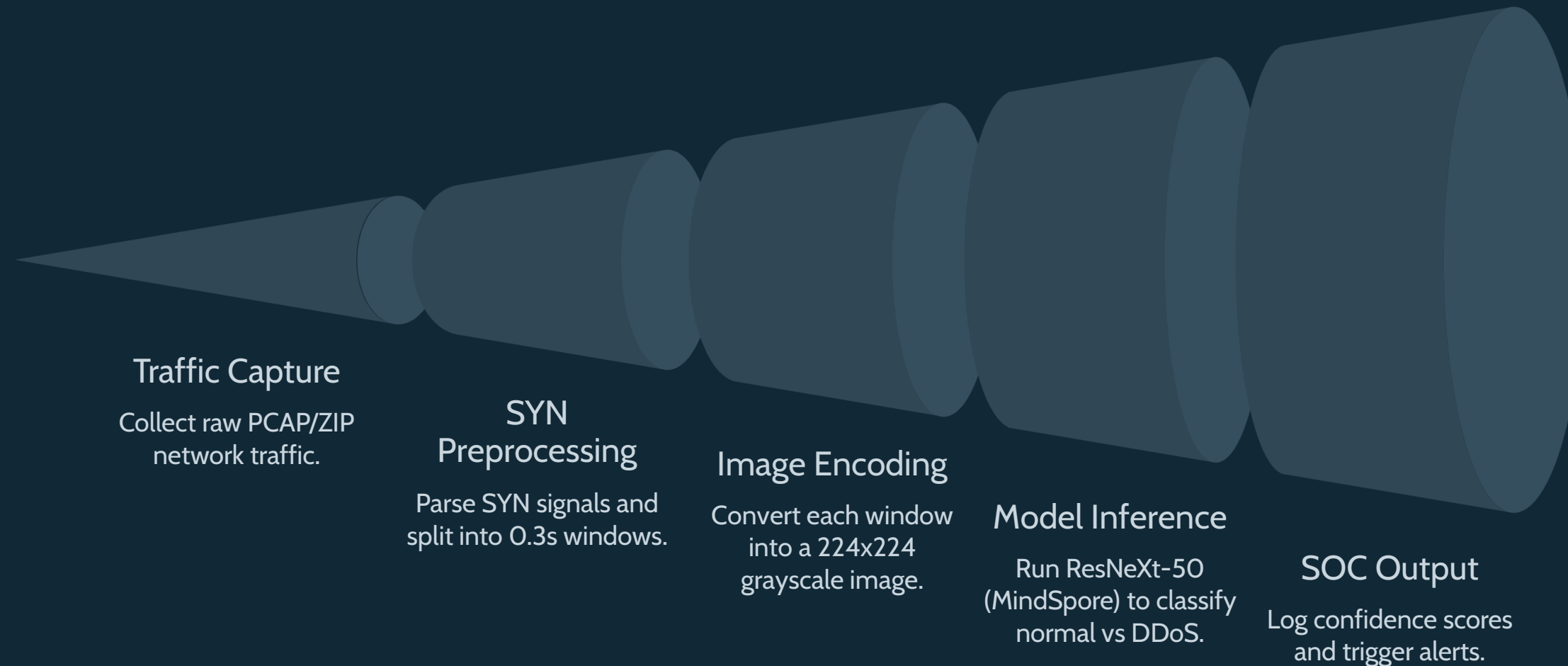
Latency & Deployment

Detection must be fast enough to trigger SOC response before damage occurs.

Why it matters: Availability protection for education, business, and public services.



Solution Snapshot



Traceable path: PCAP -> window -> image -> prediction -> alert.



What Makes SYNSheild Different



ResNeXt-50 (32×4d)

Grouped convolutions capture structured packet-image patterns better than standard CNNs.



0.3s Time Windows

Fine-grained temporal slicing preserves attack signal while keeping inference latency low.



Imbalance-Aware Training

Macro-F1 checkpointing ensures fair evaluation across imbalanced classes.



Automated End-to-End

Scripts cover data prep → training → blind test → ROC/PR → cross-domain checks.



PART B

From Value Proposition to Research Validation

The following slides follow standard research evaluation logic — methodology, dataset design, results, and reproducibility evidence.

01

Related Work & Research Gap

02

Methodology & Architecture

03

Dataset & Experimental Design

04

Results, Limitations & Next Steps



Related Work & The Research Gap

Existing Detection Families

- **Rule / Signature Systems** — fast but brittle under novel or obfuscated attack patterns
- **Handcrafted-Feature ML** — depends on expert feature engineering; limited generalization
- **Image-Based CNN Methods** — promising, but often accuracy-only and non-reproducible

The Gap SYNSheild Addresses

- Accuracy-only reporting under class imbalance
- Limited or absent blind evaluation protocols
- Fragmented, non-reproducible pipelines

SYNSheild closes these gaps with macro-F1 evaluation, blind testing, and a fully automated reproducible workflow on MindSpore.



RESEARCH DESIGN

Research Objectives & Hypothesis

Aim: Build a deployable and reproducible early SYN-flood detector.

Objectives

- Robust image representation from packet windows
- Fair training/evaluation in MindSpore
- Blind and cross-domain validation

Hypothesis

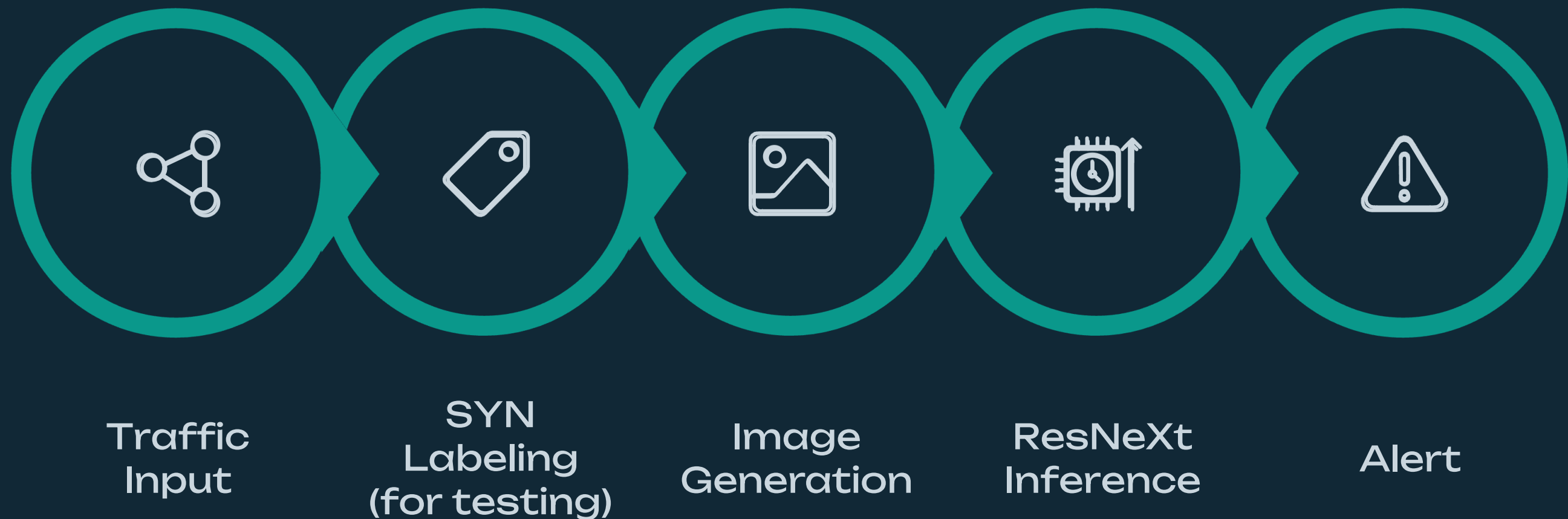
0.3s window images + ResNeXt improve practical early-detection quality.





System Model & Architecture

A deterministic, modular pipeline transforms raw network traffic into actionable detection alerts — each stage is independently testable and reproducible.

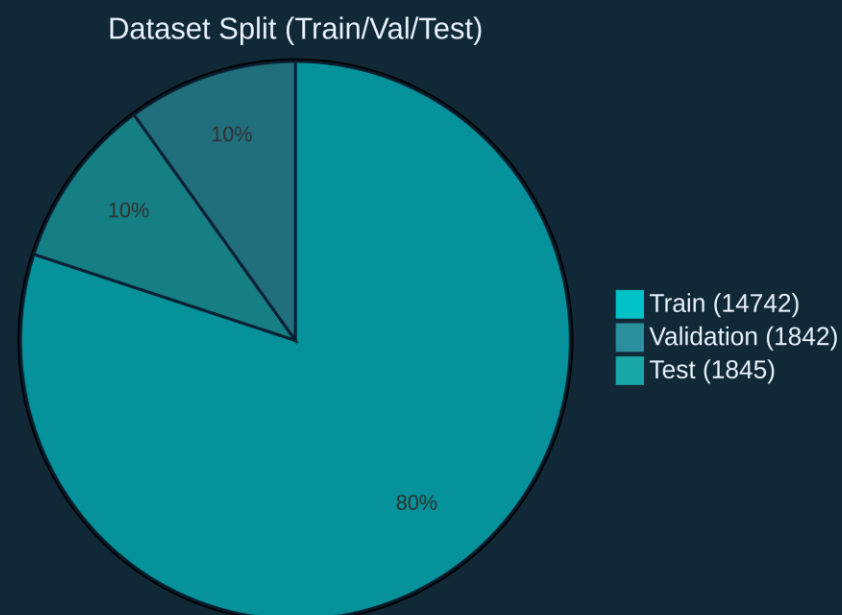


The pipeline is implemented as modular script stages, enabling isolated testing, version control, and consistent train-to-deploy representation.



METHODOLOGY

Data & Methodology



Supervised Binary Classification

Normal vs. SYN-flood attack traffic, trained and evaluated in MindSpore.

Data Sources & Splits

PCAP/ZIP datasets with train / val / test + a held-out **blind** split for domain-agnostic validation.[CICDDOS2019]

Preprocessing

SYN-oriented parsing → 0.3s
windowing → 224×224 grayscale
image conversion.

Model Design

ResNeXt-50 with imbalance-aware training and **macro-F1 checkpointing** to ensure fair evaluation.



EVALUATION

Evaluation Criteria & Experimental Setup



Primary Metrics

Accuracy and macro-F1 —
macro-F1 ensures fair scoring
across imbalanced classes.

Evidence Curves

ROC and Precision-Recall
curves provide full operating-
point visibility beyond single-
number metrics.

Controlled Protocol

Strict train/val/test separation, a balanced blind set, and checkpoint
selection by macro-F1 — no data leakage.



ENVIRONMENT

Implementation Environment & Requirements



Hardware

Training validated on RTX 3070-class GPU — consumer hardware, no specialized cluster required.



Software Stack

Python 3.10.x with CUDA-compatible stack (~11.6) for GPU-accelerated training and inference.



Frameworks

MindSpore 2.6.0 + mindcv 0.3.0 — fully open-source, Huawei-native deep learning ecosystem.

Critical Reproducibility Constraint

Train-time and deploy-time image representations must match **exactly**. Any divergence in windowing, scaling, or normalization invalidates results.



This constraint is enforced via shared preprocessing modules across all pipeline stages.

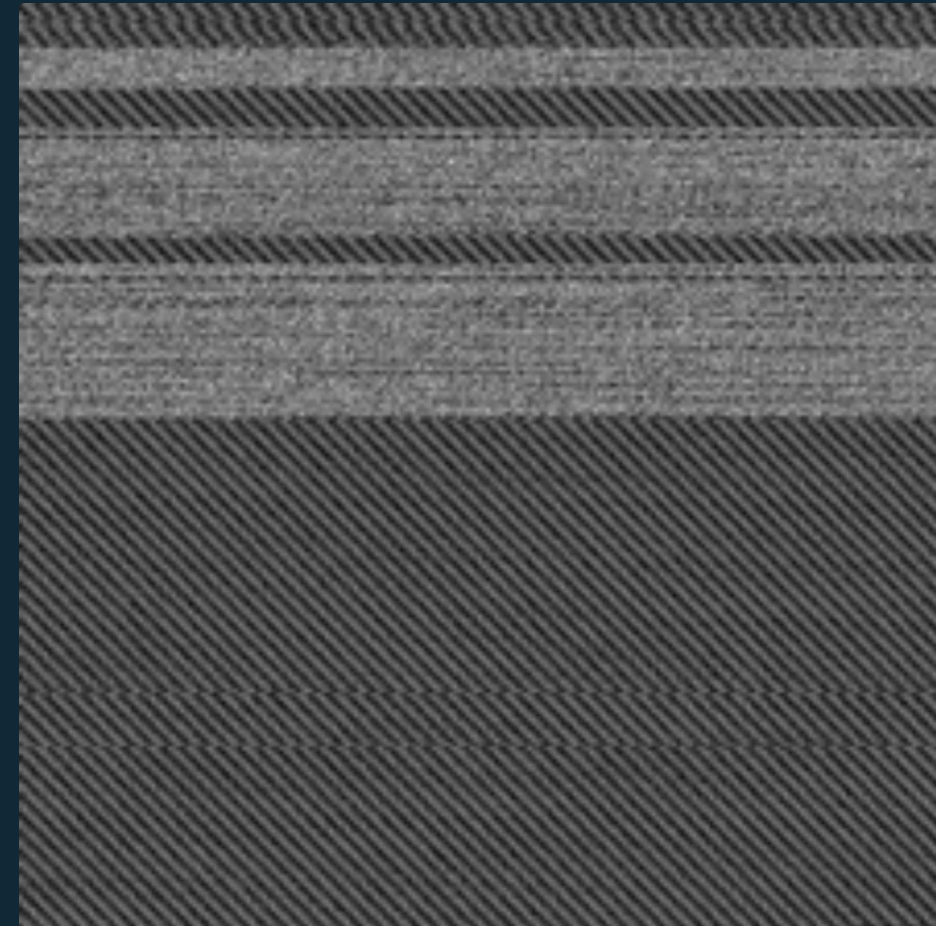
Sample Input Windows

The 0.3s window-to-image transformation exposes learnable attack structure — normal and DDoS traffic produce visually distinct texture patterns.



Normal Traffic

Sparse, low-density SYN activity. Image texture reflects benign connection patterns.



DDoS-like Traffic

High-density SYN bursts create structured, high-frequency texture — directly learnable by ResNeXt.



RESULTS

Performance Results

Checkpoint: `resnext50_32x4d_best.ckpt` (focal loss, 0.3s windows)

98.21%

Validation Accuracy

Macro-F1: 0.9399

98.10%

Test Accuracy

Macro-F1: 0.9375

98.9%

Blind Accuracy

270 balanced samples · Macro-F1:
0.9889

📄 Blind set performance exceeds test performance — strong evidence of genuine generalization, not overfitting to the training domain.

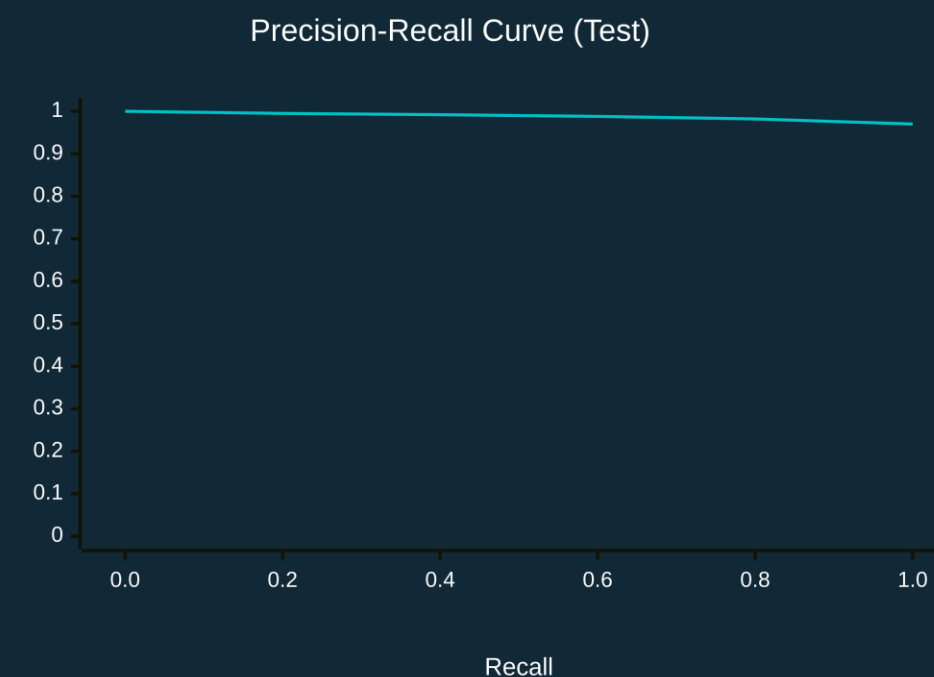
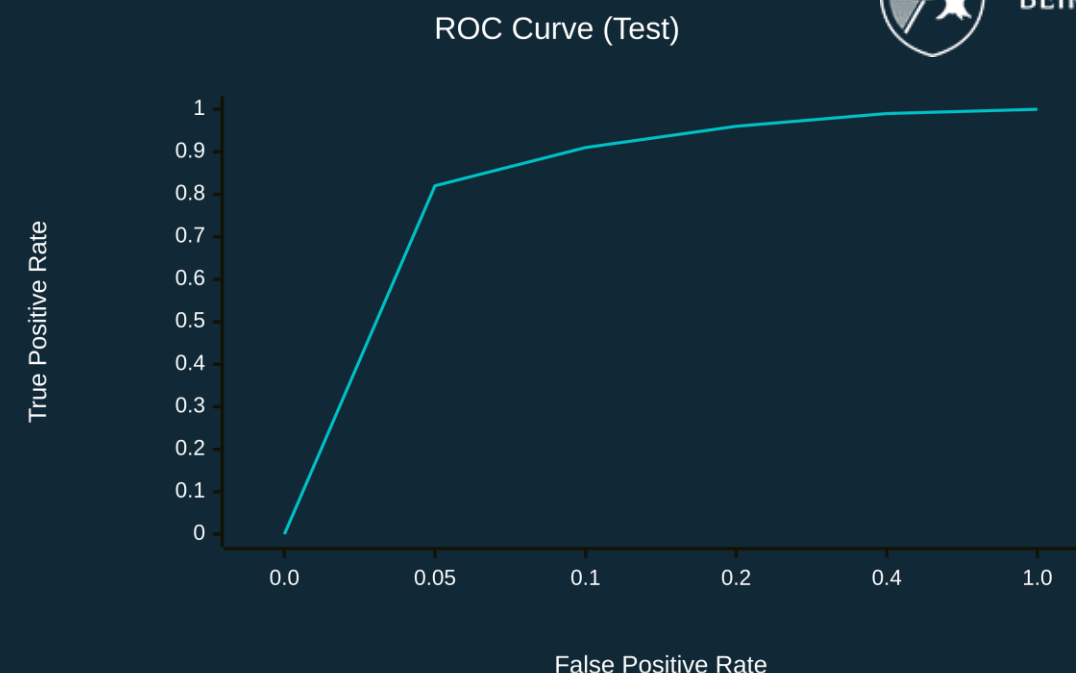
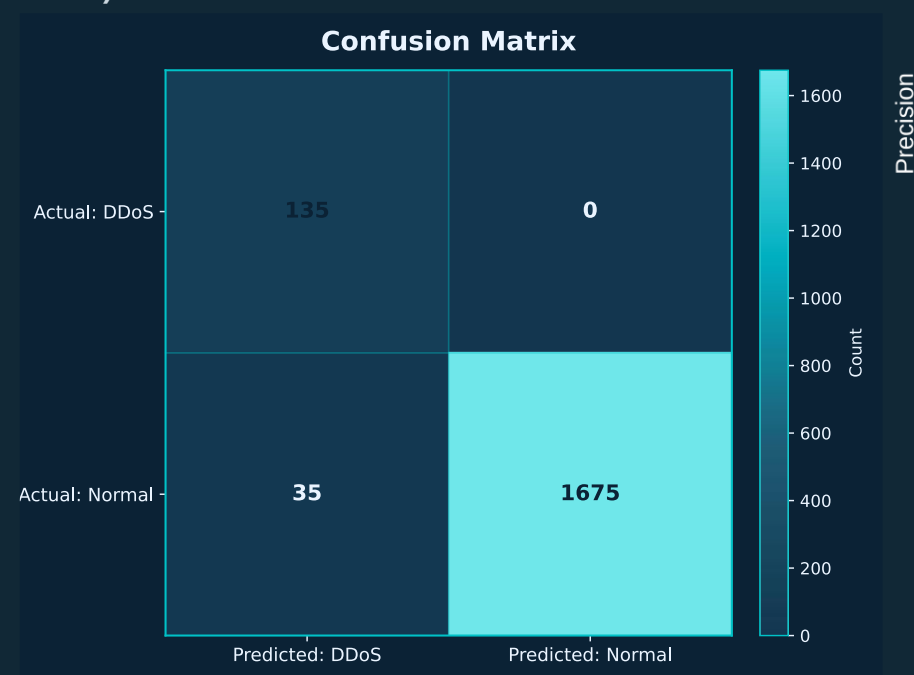


ARTIFACTS

Proof Artifacts

Existing Artifacts

- ROC and PR curve figures (validation + test)
- Model checkpoints, training logs, and markdown reports
- Full experiment traceability from raw PCAP to final metrics
- Confusion matrices (test + blind)





COMPARATIVE ANALYSIS

How We Compare

Our advantage is not a novel input concept — packet-window imagery exists in prior work. The differentiation lies in **architecture depth, fair evaluation, and end-to-end reproducibility**.

Method Family	Strength	Limitation	Our Advantage
Rule-Based	Fast, simple	Brittle under traffic variation	Better pattern learning
Handcrafted ML	Explainable	Heavy feature engineering	Less manual engineering
Prior image-CNN	Strong concept	Weaker deployment/reproducibility	MindSpore + full pipeline

→ Rule-Based Methods

Threshold-driven, brittle under adaptive attacks. No learning capacity.

→ Handcrafted-Feature ML

Dependent on expert feature engineering. Limited texture capture.

→ Prior CNN Approaches

Shallow architectures underfit complex SYN-flood texture patterns.



IMPLICATIONS

Discussion & Technical Implications

Key Findings

Strong class separation with robust macro-F1 across all splits. Blind performance (98.9%) validates genuine generalization — not domain-specific overfitting.

SOC Applicability

0.3s detection windows enable early-warning integration into SOC workflows — actionable alerts before full attack escalation.

Open Challenges

Threshold tuning per deployment environment and domain adaptation to novel traffic profiles remain important engineering considerations for production use.





Huawei Ecosystem Fit & Deployment

End-to-end deployment architecture aligned with Huawei's MindSpore ecosystem — designed for real network infrastructure, not lab-only scenarios.

Ecosystem

- MindSpore/MindCV aligned development path

Deployment Pattern

- Router/switch telemetry export
- Nearby edge/server inference node
- SOC alerts + confidence logs

Realism

- Practical pattern is telemetry + nearby inference, not CNN on every switch ASIC





Practical Deployment Pattern

Router/Switch
exports telemetry
data.

Nearby edge or server
node runs
MindSpore inference.

SOC receives
alerts with
confidence logs.



MindSpore / MindCV Alignment

Development path is fully aligned with Huawei's MindSpore framework and MindCV model library — enabling native deployment on Ascend hardware.



Design principle: Telemetry + nearby inference is the realistic pattern — CNN inference does not run on every switch ASIC.

Telemetry Export

Router and switch flow data exported via standard interfaces

Nearby Inference Node

Edge or server node performs CNN inference — not on ASIC

SOC Integration

Alerts delivered with confidence scores and audit logs



Honesty, Limitations & Roadmap

We acknowledge constraints openly. Every limitation has a mitigation path — and a timeline.

⚠️ Known Challenge

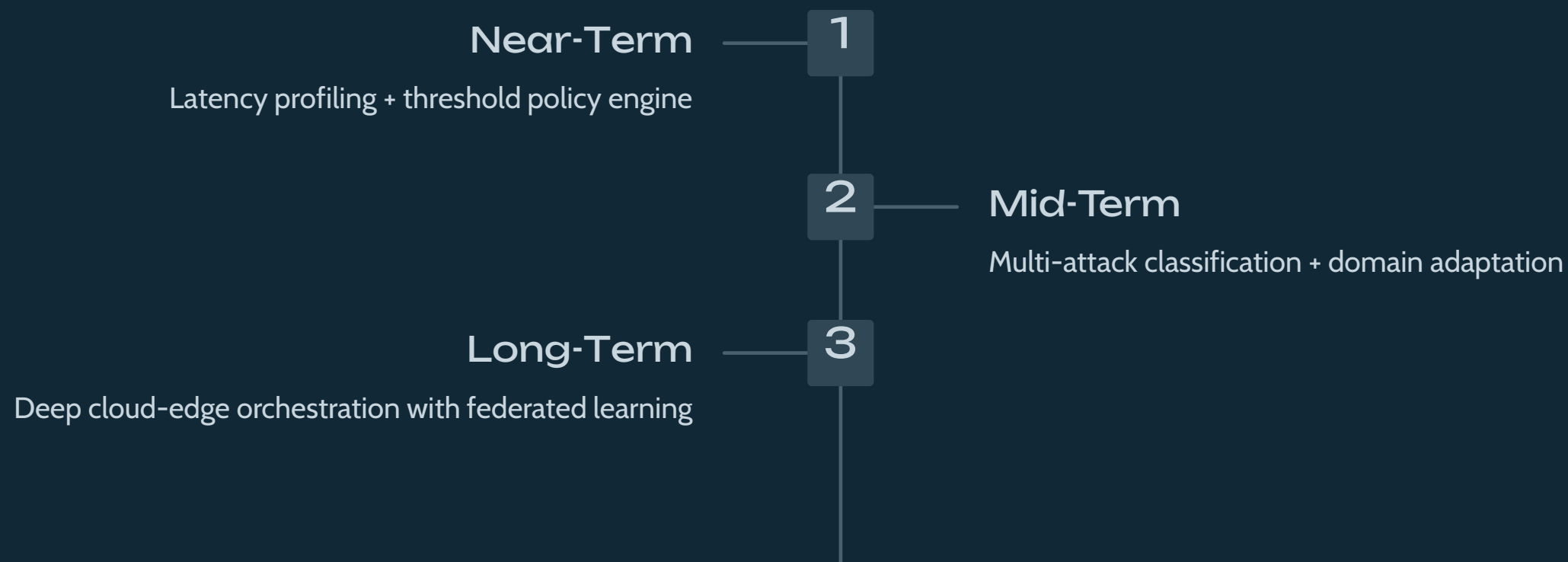
No GPU in some server environments limits real-time training capacity

✅ Mitigation

Offline GPU benchmarks + CPU-compatible inference runs validate correctness

🎯 Current Scope

Primary focus on SYN-flood class attacks — deliberate scope for rigor





CLOSING

Completeness, Contributions & Q&A

Completeness Package

Full reproducibility delivered — not just slides.

- Slides, report, and presentation scripts
- Model checkpoints and training logs
- Figures, raw logs, and repro instructions

Key Contributions

Deployable MindSpore pipeline — not a prototype

Reproducible end-to-end workflow with blind evidence

Rigorous evaluation with fair benchmarking

Innovation + Credibility + Deployment Realism

Thank you — we welcome your questions.

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

Bring digital to every person, home and
organization for a fully connected,
intelligent world.

**Copyright©2025 Huawei Technologies Co., Ltd.
All Rights Reserved.**

The information in this document may contain predictive statements including, without limitation, statements regarding the future financial and operating results, future product portfolio, new technology, etc. There are a number of factors that could cause actual results and developments to differ materially from those expressed or implied in the predictive statements. Therefore, such information is provided for reference purpose only and constitutes neither an offer nor an acceptance. Huawei may change the information at any time without notice.

